Workflow and Tech Stack

databricks-agentic-de - a real-time healthcare data platform on Databricks where the platform's agents do the work and this project builds the glue and guardrails around them

Status as of commit bfd8331, 12 Oct 2026. Repository: github.com/sumanthreddy369/databricks-agentic-de

In one paragraph

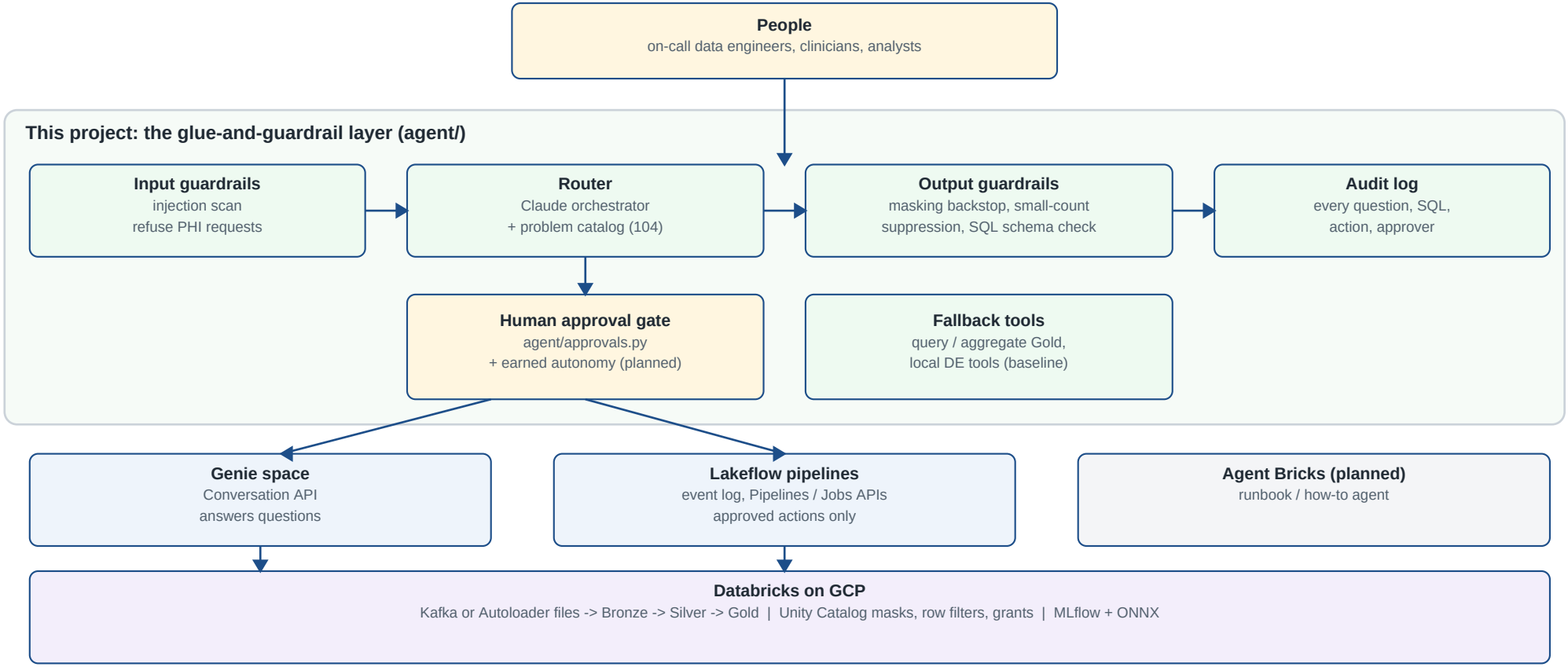
Patient events (admissions, transfers, discharges, vital signs) flow from a simulator into Databricks, through Bronze, Silver and Gold tables governed by Unity Catalog. Databricks' own agents do the data work: Genie answers analysts' questions and the pipeline APIs act on pipelines. This project is the layer around them that companies build themselves: it routes requests, checks everything going in and coming out (patient-data masking, injection scanning, small-count suppression, SQL scope), holds every change for a named human approver, and records everything. The goal is to cut routine data-engineering and analyst work as fast as is safe.

Status at a glance

Area	State
Glue-and-guardrail layer, simulator, MCP, ML, approvals, evals harness	Tested locally - 290 tests, zero network calls, passing in CI
Genie and live-workspace connections	Tested against mocked APIs only
Databricks deploy config	Validated against the real GCP trial workspace; nothing deployed yet
Pipelines, Unity Catalog SQL, Terraform	Written, never run
Real-model eval results, Genie comparison, human-work report	Not measured yet
Dataset	103,301 synthetic events generated locally; upload waiting on the catalog being created in the UI

1. Architecture

Three tiers: people at the top, this project's layer in the middle, Databricks' agents and platform underneath. Every request into the layer passes the input guardrails; every result passes the output guardrails; every action passes the approval gate; everything is audited.

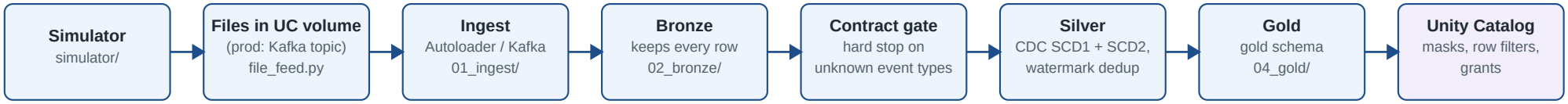


- **Platform does the work:** Genie (questions), Lakeflow pipelines and Jobs APIs (actions), Agent Bricks planned (runbooks). Databricks Assistant has no public API, so people use it directly in notebooks.
- **We build glue + guardrails:** routing, the Genie gateway, guardrails, approvals, audit, evals, reports.
- **Fallback = baseline:** when Genie isn't configured our own query/aggregate tools answer; the same tools are what Genie is measured against.

2. Workflows

W1. Data flow: source to Gold

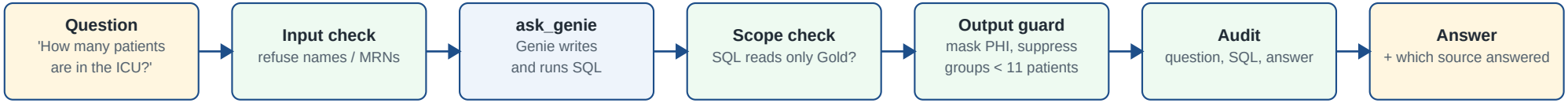
How a vital-sign reading becomes a governed Gold row. The free-trial setup uses files instead of Kafka; both feed identical columns to Bronze.



Step	What happens	Rules	Code	Status
Generate	10 min of events for 200 patients, stamped on a simulated clock	No future timestamps; one file per simulated minute	simulator/file_feed.py	Tested locally
Ingest	Autoloader reads JSON lines (or Kafka); parses with the contract schema	Malformed messages kept in _corrupt_record, never dropped	pipeline/01_ingest/kafka_patient_events.py	Validated
Bronze	Flatten payload, add ingestion time	Never drops a row; warn-only checks	pipeline/02_bronze/	Validated
Silver	Contract gate, then encounters (current + history), patients, vitals	Unknown event type fails the pipeline; impossible vitals (0 HR, SpO2 > 100) dropped; clinical alarms kept	pipeline/03_silver/	Validated
Gold	dim_patients, fct_encounters, fct_encounter_history, fct_vitals, dim_providers, live vitals by unit	Readings attributed to the unit at reading time; UNASSIGNED instead of dropping	pipeline/04_gold/	Validated
Govern	Masks on full_name / mrn, unit row filters, SELECT-only grant for the agent	Agent never in phi_unmasked	governance/05_unity_catalog/	Never run

W2. Analyst question

A clinician asks a plain-English question. Genie answers; our layer decides whether the question may be asked and what of the answer may be shown.



Step	What happens	If it goes wrong	Code	Status
Input check	Requests for patient identifiers are refused before anything is sent	Refusal returned; nothing reaches Genie	agent/tools/genie.py	Mock-tested
Genie	Conversation API: start, poll, fetch query result	Failure / timeout -> tool error; not configured -> fallback	agent/databricks_client.py	Mock-tested
Fallback	aggregate_gold_table computes counts/averages in SQL; query_gold_table returns masked rows	Never counts capped rows; empty result = fixed refusal	agent/tools/data_query.py	Tested locally
Output guard	Masked columns redacted, leaked values removed from text, small counts suppressed, rows capped	Result withheld if SQL left Gold	agent/tools/genie.py	Mock-tested
Audit	One append-only line per tool call, masked	-	agent/orchestrator.py	Tested locally

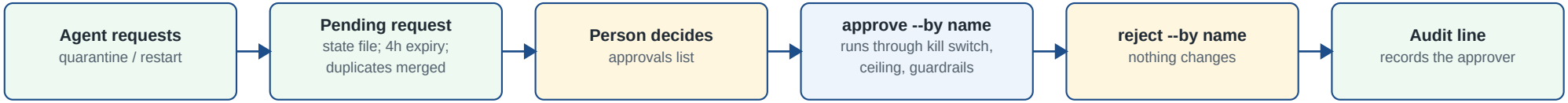
W3. Pipeline incident

Every 15 minutes the healthcheck job asks the agent to check the pipelines. The problem catalog decides how far the agent may go.

Step	What happens	Rules	Code	Status
Check	Expectation failures, job states, schema drift (local state file or Pipelines API)	Read-only	agent/tools/pipeline_health.py, _live.py	Tested locally / mock-tested
Match	Symptom matched to one of 104 catalog problems	Most restrictive level wins; no match = new problem -> escalate	agent/tools/problem_catalog.py	Tested locally
Escalate	Contract breaks, data loss, schema drift go to a person	Never auto-fixed	notify_and_page	Tested locally
Request fix	Quarantine / restart become pending requests	Default: approval required	agent/orchestrator.py	Tested locally
Exit code	0 healthy, 1 agent down, 2 escalated, 3 awaiting approval	Non-zero fails the job -> failure email to a person	agent/healthcheck.py	Tested locally

W4. Human approval lifecycle

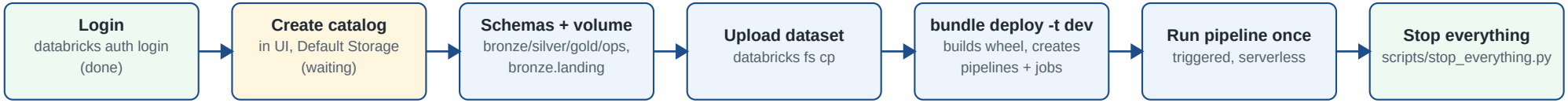
How a requested fix becomes an executed one - or doesn't. The model is never offered an approve or reject tool.



Command	Effect	Notes	Code	Status
python -m agent.approvals list	Shows pending, unexpired requests	Read-only	agent/approvals.py	Tested locally
... approve --by "Name"	Executes the action now	Fails if expired, kill switch on, or ceiling reached	agent/approvals.py	Tested locally
... reject --by "Name" --reason ...	Closes the request	Audited as approval_rejected	agent/approvals.py	Tested locally
remediation_approval: auto	State-file opt-out for local demos	Real deployments start with 'required'	agent/healthcheck.py	Tested locally
Earned autonomy	Proven L2 fix classes promoted to auto; demoted on first failure	Destructive, permission, prod: always human	docs/plan.md	Planned

W5. Deploy and run on the free trial

From laptop to a running pipeline on the GCP trial workspace, using serverless compute only (trial credits, no VMs in the GCP project).

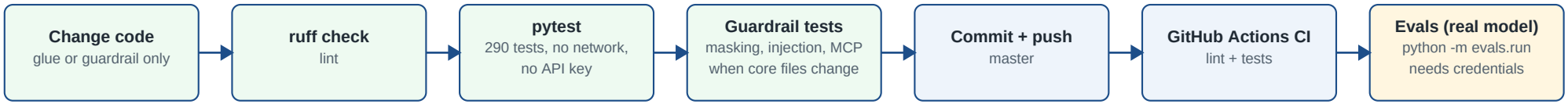


Step	Command	Cost	Notes	Status
Validate	databricks bundle validate -t dev	Free	Passes on the real workspace	Validated
Generate data	uv run python -m simulator.file_feed	Free (local)	103,301 events ready, 34 MB, git-ignored	Tested locally
Upload	databricks fs cp -r data/landing/... dbfs:/Volumes/...	Storage only (cents)	After the catalog exists	Planned
Deploy	databricks bundle deploy -t dev	Free until something runs	Dev mode pauses schedules	Planned
Run	databricks bundle run streaming_patient_pipeline	Serverless DBUs from trial credits	Triggered, not continuous	Planned
Stop	uv run python scripts/stop_everything.py	-	--dry-run lists; --destroy removes deployed resources	Validated (dry run)

- Only the account owner can guarantee $0 after the trial: set a GCP budget alert and cancel the Databricks subscription in GCP Marketplace (or delete the workspace) before the trial ends.

W6. Development and quality workflow

How a change gets in. The repo's credibility rests on an honest split between tested-locally, validated, mock-tested and never-run.



Practice	Why	Where	Enforced by	Status
No real secrets in the repo	Public repo	.gitignore, agent/secrets.py	Review; Secret Manager	Tested locally
Status labels in every doc	Never claim untested work works	README feature status	Review	Tested locally
One file per concern in tests	Guarantees stay provable	tests/	CI	Tested locally
Live paths mock-tested	No network in CI	tests/conftest.py clears DATABRICKS_*	conftest fixture	Tested locally
Problem catalog = PDF source	Docs can't drift from what the agent knows	agent/knowledge/, scripts/	Generator reads the YAML	Tested locally

3. Tech stack

Versions are the ones pinned in uv.lock at this commit. Status colours: green tested locally; yellow validated or mock-tested; red never run; grey planned.

Layer	Technology	Version	What it does here	Repo path	Status
Language / tooling	Python, uv, ruff, pytest	Python >=3.11; ruff 0.16.8; pytest 9.1.1	Package management (uv only), lint, 290 tests with zero network calls	pyproject.toml, uv.lock, tests/	Tested locally
Data source	Patient-event simulator (Synthea / MIMIC-IV-shaped, synthetic)	-	Admit / transfer / discharge + 6 vitals; Kafka publisher or JSON-lines files	simulator/	Tested locally
Transport (prod)	Kafka / Redpanda	confluent-kafka 2.15.1	Production event stream; local broker via Docker	docker-compose.yml, simulator/producer.py	Tested locally
Transport (free trial)	Autoloader over a Unity Catalog volume	-	Reads JSON-lines files; no extra service, no extra cost	simulator/file_feed.py, pipeline/01_ingest/	Validated
Pipeline	Lakeflow Declarative Pipelines (formerly DLT), serverless	-	Bronze -> contract gate -> Silver (CDC SCD1 + SCD2, watermark dedup) -> Gold	pipeline/, resources/dlt_pipeline.yml	Validated - bundle passes on the real workspace; never run
Contract	Pydantic + shared constants	pydantic 2.13.5	Single source of truth: schema, masked columns, physical vital limits, min cell size	common/contracts.py	Tested locally
Governance	Unity Catalog	-	Column masks, row filters, grants, landing + ops volumes	governance/05_unity_catalog/	Never run
Storage / query (live)	Delta tables, Databricks SQL warehouse (serverless)	-	Gold tables; Statement Execution API for queries	agent/databricks_client.py	Mock-tested
Storage / query (local)	DuckDB	duckdb 1.5.5	Local Gold stand-in for tests and demos	data/seed/gold_seed.sql	Tested locally
Platform agent (questions)	Databricks Genie	Conversation API	Writes and runs SQL for analyst questions	agent/tools/genie.py	Mock-tested
Platform agent (runbooks)	Databricks Agent Bricks	-	Planned knowledge agent over the problem catalog	docs/comparisons/	Planned
Router / LLM	Anthropic Claude via the Anthropic SDK	anthropic 1.7.0	Mode routing and tool loop; no agent framework	agent/llm.py, agent/orchestrator.py	Tested locally (scripted model)
Tool protocol	Model Context Protocol (MCP)	mcp 2.2.0	DE tools served over stdio to the orchestrator	mcp_server/server.py, agent/mcp_bridge.py	Tested locally
Knowledge	Problem catalog (YAML) + lookup_problem	pyyaml 6.0.3	104 real-time DE/DA problems with expected response and autonomy level	agent/knowledge/, agent/tools/problem_catalog.py	Tested locally
ML	scikit-learn IsolationForest -> MLflow -> ONNX	scikit-learn 1.9.1, mlflow 3.16.1, skl2onnx 1.20.0, onnxruntime 1.30.0	Joint vital-sign anomaly scoring at inference without training deps	ml/, agent/tools/anomaly_score.py	Tested locally

Layer	Technology	Version	What it does here	Repo path	Status
Orchestration	Databricks Workflows + Asset Bundles	Databricks CLI v1.19.0	Deploy config, hourly roster job, 15-minute agent healthcheck job	databricks.yml, resources/workflows.yml, agent/healthcheck.py	Validated - never deployed
Human approval	Approvals CLI over the state file	-	Queued quarantines/restarts; named approver; 4h expiry	agent/approvals.py	Tested locally
Secrets	GCP Secret Manager	google-cloud-secret-manager 2.30.0	Tokens/keys never in files; no-op unless configured	agent/secrets.py	Tested locally (no-op path)
Observability	OpenTelemetry, Langfuse, structlog	opentelemetry-sdk 1.45.0, langfuse 4.15.6, structlog 26.1.0	Spans, LLM traces, structured logs; no-op unless configured	agent/otel.py, agent/llm.py	Tested locally (no-op path)
HTTP / retries	httpx, tenacity	httpx 0.28.1, tenacity 9.1.4	Databricks REST client; retry on transient errors	agent/databricks_client.py, agent/llm.py	Tested locally
Infrastructure	Terraform, Cloud Run (GCP)	Terraform >=1.5	GCS bucket, service account, Pub/Sub, Secret Manager	infra/	Never run
CI	GitHub Actions	-	Lint + tests on every push; bundle validate when secrets exist	.github/workflows/	Tested locally - passing on GitHub
Evals	Scenario harness with deterministic graders	-	14 scenarios graded on actions, citations, answers	evals/	Tested locally (scripted model); no real-model run yet
Cost control	stop_everything script	-	Cancel runs, pause schedules, stop pipelines/warehouses, terminate clusters	scripts/stop_everything.py	Validated - dry run on the workspace

4. Guardrails

Enforced in code on every request, not just asked for in a prompt.

Guardrail	What it stops	Where	Proof
PHI masking before any model	Names / MRNs reaching the LLM, transcript, or audit log	agent/tools/governance_guard.py, orchestrator _dispatch	tests/test_masking_guard.py
PHI-request refusal	Asking Genie (or the fallback) for identifiers	agent/tools/genie.py; data_query masked-filter refusal	tests/test_genie_gateway.py, test_data_query_live.py
Masking backstop on Genie output	A misconfigured UC mask leaking values, incl. into Genie's text	agent/tools/genie.py	tests/test_genie_gateway.py
SQL schema check	Genie results from tables outside Gold	agent/tools/genie.py	tests/test_genie_gateway.py
Small-cell suppression	Counts / averages over fewer than 11 patients	aggregate_gold_table (exact), ask_genie (heuristic)	tests/test_aggregate_tool.py
Prompt-injection scan	Instructions hidden in free text (e.g. nurse notes)	governance_guard.scan_for_injection on every tool result	tests/test_prompt_injection_guard.py
Human approval	The agent changing pipelines on its own	agent/approvals.py, orchestrator	tests/test_human_approval.py
Kill switch + escalation ceiling	Remediation loops; fixing past a hard stop	orchestrator	tests/test_kill_switch.py, test_escalation_ceiling.py
Audit log	Untraceable actions	orchestrator _append_audit_log	tests/test_audit_trail.py
Graceful degradation	LLM outage crashing callers or looking green	orchestrator.handle, healthcheck exit 1	tests/test_healthcheck.py
Size caps, groundedness	Huge results; speculating on empty data	orchestrator, data_query	tests/test_groundedness_refusal.py

5. Where everything lives

Folder / file	Contents
agent/	The glue-and-guardrail layer: orchestrator, LLM wrapper, prompts, approvals, healthcheck, Databricks client, MCP bridge
agent/tools/	genie (Genie gateway), data_query (query + aggregate), pipeline_health (+ _live), problem_catalog, governance_guard, anomaly_score, knowledge_search
agent/knowledge/problem_catalog.yaml	104 real-time problems: how they show up, expected response, autonomy level, status
mcp_server/	MCP server exposing the DE tools
common/contracts.py	Event schema, masked columns, physical vital limits, Gold schema, min cell size
simulator/	Event generators, Kafka producer, JSON-lines file feed, roster feed, chaos injectors
pipeline/	Lakeflow pipeline code: 01_ingest, 02_bronze, 03_silver, 04_gold, common/schemas.py
governance/05_unity_catalog/	Catalog, schemas, volumes, grants, masks, row filters
ml/	Anomaly model training and the committed ONNX file
databricks.yml, resources/	Asset Bundle: dev target (trial workspace), pipelines, jobs
evals/	Scenarios, graders, harness, real-model runner
scripts/	stop_everything.py, build_problem_catalog.py
infra/	Terraform and Cloud Run (never applied)
docs/	plan.md, architecture.md, stack_flow.md, builder study, problem catalog PDF, comparison plans
tests/	290 tests, one file per concern

Pull it

- `gh repo clone sumanthreddy369/databricks-agentic-de`
- `uv sync --extra dev` then `uv run python -m pytest`

6. What's next

#	Step	Needs	Outcome
1	Create catalog healthcare_agentic_de (UI), then schemas, volume, upload dataset	You: 30 seconds in the UI	Data in the workspace
2	Deploy to dev and run the pipeline once; fix what the first run surfaces	Approval to deploy	First real pipeline run
3	Genie space over Gold; point ask_genie at it	Workspace with Gold data	Genie behind our guardrails, live
4	Metrics layer: one definitions file -> UC metric views + Genie instructions	-	Genie and fallback share definitions
5	Human-work report from the audit log and approvals	-	Baseline for 'less human work'
6	Earned-autonomy policy	Report data	Approvals that retire themselves safely
7	Real-model evals and the Genie comparison	Anthropic credentials; Genie space	Measured pass rates and comparison numbers
8	Before trial ends: stop everything; cancel subscription	Trial end date; you in GCP Marketplace	No charges